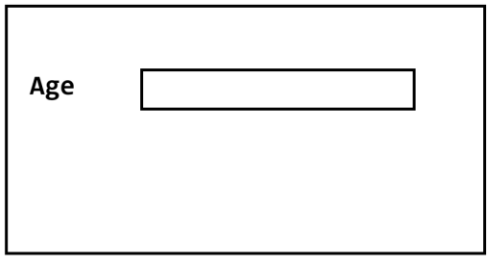
Test Case Design Technique/ Test Design Technique/ Test Data Design Technique

#. Used to prepare data for testing.

```
Age Criteria to take a loan.

Age Limit 28-58


1. Reduce the data.
2. More Coverage
```



Types: -

1. Equivalence class Partitioning (ECP)

2. Boundary Value Analysis (BVA)

3. Decision Table based testing.

4. State Transition

5. Error Guessing

1. Equivalence class Partitioning (ECP)

#. Partition data into various classes and we can select data according to class then test.

It reduce the number of test-cases and saves time for testing.

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```
Enter a number [          ]
```

* Allow Digits from 1-600

```
Normal Test Data
1
2
3
4
--
--
.
.
.
600
```

```
Divide values into Equivalence
Classes

-100 to 0 : -40 ( Invalid)
1 to 100 : 33 ( Valid)
101 to 200 : 150 (Valid)
201 to 300 : 299 (Valid)
301 to 400 : 355 (Valid)
401 to 500 : 450 (Valid)
501 to 600 : 555 (Valid)
601 to 700 : 640 (Invalid)
```

```
Test Data using ECP

-40
33
150
299
355
450
555
640
```

* Allow only alphabets

```
Name: [          ]
```

```
Divide values into Equivalence Classes

A to Z : XYZ (Valid)
a to z : pqr (Valid)
Special Character  @#$ ( Invalid)
Spaces : Xy p  (Invalid)
Numbers : 1234 (Invalid)
```

```
Test Data using ECP
------------
XYZ
pqr
@#$
Xy p
1234
```

Generate a Utility (Electric)Bill for Customer

Customers

Residential

Commercial

Industrial

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Company
Government

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1. Regular Bill (30 Days)

2. Proration Bill (Less No of Days) (15 Days)

3. Proration Bill (More No of Days) (35 Days)

```
2. Boundary Value Analysis (BVA)
#. BVA technique used to check Boundaries of the input.

      Allow Digit from 18 to 50
Age:  [                ]

Min= 18(Pass)                Max=50(Pass)
Min-1= 17 (Fail)             Max-1=49(Pass)
Min+1=19 (Pass)              Max+1=51(Fail)

Input Domain Testing:-
#. The Values will be verified in the text box/input box
#. We use ECP and BVA techniques
```

```
Decision Table:-

#. This Technique will be used if we have more conditions and corresponding actions.
#. In Decision Table Technique , we deal with combination of inputs.
```

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Scenario:
Transfer Money Online.

Conditions:-
- Account is in Approved status.
- OTP (One Time Password) matched
- Sufficient Fund/Money in the account

Actions:-

- Transfer Money
- Show Message as insufficient amount
- Block the account incase of suspicious transaction.

A	B	C	D	E	F	G
		TC1	TC2	TC3	TC4	TC5
Condition 1	Account should be Active State.	TRUE	TRUE	TRUE	TRUE	FALSE
Condiiton 2	OTP (One Time Password) match.	TRUE	TRUE	FALSE	FALSE	X
Condiiton 3	Sufficient Money in the bank.	TRUE	FALSE	TRUE	FALSE	X
Action 1	Transfer Money Success.	Execute				
Action 2	Show a Message as InSufficient amount		Exceute			
Action 3	Block the transaction in case of suspicious transaction.			Execute	Execute	X

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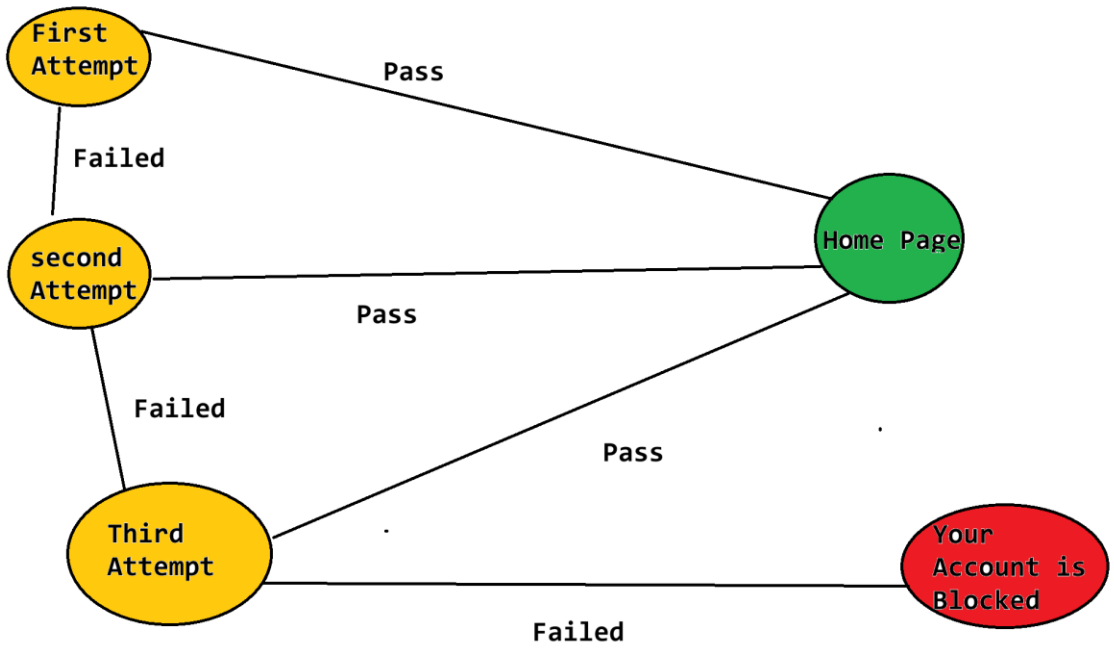
State Transition:-

#. In State Transition Technique changes in input condition changes the state of the application.
#. The Test Engineer can perform this action by entering various input condition in sequence.
#. In State Transition technique, the testing team provides positive as well as negative input test values for evaluating the system behavior.

Login Module



STATE	LOGIN	CORRECT CREDENTIALS	INCORRECT CREDENTIALS
S1	First Attempt	S4	S2
S2	Second Attempt	S4	S3
S3	Third Attempt	S4	S5
S4	Home Page		
S5	Account Locked		

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Error Guessing:-

#. Error Guessing is one of the testing technique used to find bugs in the application based on the testers prior experience.

#. In Error Guessing we don't follow any specific rules.

#. It depends on Test Engineer Analytical Skills and Experience.

Example:-

#. try to submit form without entering values.

#. try entering invalid input such as in alphabets in the numeric text field.

Static Testing Technique

1. Review

2. Walkthrough

3. Inspection

Dynamic Testing Techniques

1. Unit Testing

2. Functional Testing

3. Integration Testing

4. System Testing

5. UAT (User Acceptance Testing)

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Static Testing Technique

1. Review

2. Walkthrough

3. Inspection

1. Review:-

#. Conduct on documents to ensure correctness and completeness.

#. It is a formal review.

#. Requirement review

#. Design review

#. Code Review

#. Test Plan Review

#. Test Cases Review.

2. Walkthrough:-

#. It is an informal review.

#. Author reads the document and discuss with the peers.

#. It is not pre-planned and can be done whenever required.

#. Also walkthrough does not have MOM(Minutes of the Meeting)

Inspection:-

#. Its a formal review type.

#. Inspection will have proper schedule which will be intimated via email to the concerned people. (Manager/Lead)

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